Project Name: SHOBKAAJ

Course: PROGRAMMING IN PYTHON [A]

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Track Name

Software Engineering / CLI Application System

Category: Job Marketplace System using Python (Rich + JSON)

Project Overview

This project is a command-line based job marketplace platform where clients can post jobs and workers can apply for them. The system is implemented using Python and uses JSON for data storage. The Rich library is used to provide a visually enhanced terminal interface.

Setup & Run Instructions

1. Clone the Repository

Download the project from GitHub using the following command:

git clone **https://github.com/mahito-0/Python_Project_Mid**

2. Navigate to Project Directory

Move into the project folder:

cd Python_Project_Mid

3. (Optional) Install Dependencies

If running in a fresh environment, install required packages:

pip install -r requirements.txt

Note: This project mainly uses built-in Python libraries. The Rich library may be required for styled terminal output.

4. Run the Application

Execute the main program:

```
python main.py
```

5. Using the Platform

After running the program, the Shobkaaj Platform menu will appear in the terminal.

```
=== Shobkaaj Platform ===

1. Register as Worker
2. Register as Client
3. Manage Jobs
4. Apply to a Job
5. View Applications
6. Exit
```

Users can interact with the system by entering the corresponding number and following the on-screen instructions.

Functionalities

The application provides a fully interactive Command Line Interface (CLI) that allows users to perform various operations related to a job marketplace system. The key features and functionalities of the system are described below:

Worker Registration

The system allows users to register as workers by providing detailed information including ID, name, skills, experience, location, hourly rate, and current availability status. This information is stored persistently for future interactions.

Client Registration

Users can register as clients or companies by entering details such as ID, name, company name, location, offered rate, and status. This enables clients to post and manage job listings within the platform.

Job Management (CRUD Operations)

The application supports complete job lifecycle management through Create, Read, Update, and Delete (CRUD) operations:

- Post a Job: Clients can create new job listings by specifying job title, description, location, and payment rate.
- View Open Jobs: Users can view all available jobs that are currently marked as "Open" in a structured and formatted output.
- Update a Job: Existing job details can be modified dynamically. The system ensures proper input validation to maintain data integrity.
- Delete a Job: Jobs can be safely removed from the system using their unique Job ID.

Job Application System

Workers can apply to available jobs by providing their Worker ID along with the corresponding Job ID. This creates a link between workers and job postings.

Application Tracking

The system allows users to view all submitted job applications, enabling tracking and management of worker-client interactions.

Persistent Data Storage

All system data, including workers, clients, jobs, and applications, is stored in JSON files within the data/ directory. This ensures data persistence, meaning that all information is retained even after the application is closed and restarted.

Styled CLI Interface

The application utilizes a styled command-line interface to enhance user experience, presenting information in a clear and organized manner.

Main Menu Interface

```
(venv) PS C:\Users\mahmu\Documents\GitHub\Python_Project_Mid> python main.py

╭─ === Shobkaaj Platform === ─╮
│ 1. Register as Worker      │
│ 2. Register as Client      │
│ 3. Manage Jobs             │
│ 4. Apply to a Job          │
│ 5. View Applications       │
│ 6. Exit                    │
╰────────────────────────────╯
Choose an option (1-6):
```

This screenshot shows the main menu of the Shobkaaj platform, where users can select options such as registration, job management, applying for jobs, and viewing applications.

Worker and Client Registration

```
(venv) PS C:\Users\mahmu\Documents\GitHub\Python_Project_Mid> python main.py

╭─ === Shobkaaj Platform === ─╮
│ 1. Register as Worker      │
│ 2. Register as Client      │
│ 3. Manage Jobs             │
│ 4. Apply to a Job          │
│ 5. View Applications       │
│ 6. Exit                    │
╰────────────────────────────╯
Choose an option (1-6):
```

This screenshot represents the registration section, allowing users to sign up either as a worker or as a client in the system.

Manage Jobs

```
--- Manage Jobs ---
1. Post a Job
2. View Open Jobs
3. Update a Job
4. Delete a Job
5. Back to Main Menu

Choose an option (1-5):
```

This screenshot displays the job management menu, where clients can post, view, update, or delete job listings.

View Open Jobs

Open Jobs

Job ID	Job Title	Description	Location	Rate	Status
1	Website Built	Need a website for work	Dhaka	$1000.0	Open
2	Homework	Need a person to do my home work	Dhaka	$2000.0	Open

```
--- Manage Jobs ---
1. Post a Job
2. View Open Jobs
3. Update a Job
4. Delete a Job
5. Back to Main Menu

Choose an option (1-5):
```

This screenshot shows a table of available jobs, including job ID, title, description, location, rate, and current status.

Apply to a Job

```
(venv) PS C:\Users\mahmu\Documents\GitHub\Python_Project_Mid> python main.py

=== Shobkaaj Platform ===
1. Register as Worker
2. Register as Client
3. Manage Jobs
4. Apply to a Job
5. View Applications
6. Exit

Choose an option (1-6):
```

This screenshot represents the option where by giving input 4 workers can apply for available jobs through the system.

View Applications

Applications

App ID	Job ID	Worker ID	Status
1	2	1	open
2	1	1	open

```
=== Shobkaaj Platform ===
1. Register as Worker
2. Register as Client
3. Manage Jobs
4. Apply to a Job
5. View Applications
6. Exit

Choose an option (1-6):
```

This screenshot displays submitted job applications, showing application ID, job ID, worker ID, and application status.